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Changes in metacognitions and executive functions during mindfulness and acceptance-based intervention among individuals with anxiety disorders: A randomized waitlist-controlled trial

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ABSTRACT

Introduction: This study illustrates how mindfulness and acceptance-based intervention (MABI) can change metacognitions and executive functions among individuals with anxiety disorders.

Methods: Forty-five Iranian individuals (mean [SD] age, 24.60 [4.94] years; 51.1 % female) with comorbid anxiety disorders participated in a randomized, waitlist-controlled trial of weekly group sessions of either MABI or waitlist control (WLC). Primary and process of change outcomes were assessed at baseline, post-intervention, and 1-month follow-up.

Results: Significant changes were observed over time on anxiety symptoms (Cohen $d = 1.04$, 95% CI [0.40 ± 1.65]), as well as on metacognitive beliefs (Cohen $d = 1.04$, 95% CI [0.40 ± 1.64]), executive functions (Cohen $d = 0.91$, 95% CI [0.28 ± 1.51]), and mindfulness facets (Cohen d = 0.97, 95% CI [0.34 ± 1.58]), in favor of MABI over WLC.

Conclusions: Overall, findings add to the knowledge in the field and provide cross-cultural evidence in support of MABIs as interventions that target anxiety symptoms, metacognitive beliefs and executive functions in a non-Western culture.

1. Introduction

Internationally, anxiety disorders are the most commonly experienced class of mental health problems (American Psychiatric Association, 2022; Baxter, Scott, Vos, & Whiteford, 2013; Kessler, Petukhova, Sampson, Zaslavsky, & Wittchen, 2012). They can be debilitating and costly (Konnopka & König, 2020) and are often comorbid with other psychiatric disorders (Kaufman & Charney, 2000; Kessler et al., 2012). People living in low- and middle-income countries (LMICs) are at higher risk to develop anxiety disorders due to the impact of socioeconomic and environmental stressors including income inequality, unemployment, poverty, rapid socioeconomic changes, social instability, traumas, cumulative stresses, uncontrollability and unpredictability of situations (Alonso et al., 2018; Yatham, Sivathasan, Yoon, da Silva, & Ravindran, 2018). For example, in a recent Iranian Mental Health Survey (IranMHS), anxiety disorders were the most common disorders with a 12-month prevalence rate of 15.6%, with higher prevalence in females (19.4%) than males (12.0%), and significantly more common in unemployed individuals and people with low socioeconomic conditions living in urban areas (Hajeb et al., 2018).

Despite the high prevalence and public health burden of anxiety disorders in LMICs the majority of people suffering from common mental disorders in these countries do not receive psychotherapy and, of those who do, only a relative minority receive evidence-based treatment (Alonso et al., 2018; Cuijpers, Eylem, Karyotaki, Zhou, & Sijbrandij, 2019; Fu, Burger, Arjadi, & Bockting, 2020; Knipe et al., 2019; Singla et al., 2017). A limited budget and grants for mental health, low mental health literacy, mental health stigma, poor access to mental health services, availability of training and supervision, as well as the small number of available mental health professionals outlined as the notable barriers contributing to the inaccessibility of mental health care in LMICs (Cuijpers et al., 2019; Evans-Lacko et al., 2018; Sarikhani,

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Bastani, Rafiee, Kavosi, & Ravangard, 2021; Zemestani, Hosseini, Petersen, & Twohig, 2022). Although there are no specific data available regarding the mental health treatment-seeking among Iranian university students, the rate of psychological treatment seeking is very low in this population due to the stigma, low mental health literacy, and cultural beliefs (Ghadirian & Sayarifard, 2019; Mahmoodi et al., 2022). Given high prevalence of anxiety disorders and barriers to receiving mental health services in LMICs, there is a growing need to develop and expand effective evidence-based treatments for people in these countries.

Anxiety disorders usually have a chronic course and, without treatment, are linked to adverse long-term outcomes (Revicki et al., 2012), indicating the need for effective interventions. Several evidence-based interventions are available that effectively target cognitive and emotional variables associated with anxiety disorders. The past two decades have witnessed the growth of mindfulness and acceptance-based interventions (MABIs) as a generation of rapidly emerging therapies that belong to the family of third wave and process-based approaches (Ciarrochi, Sahdra, Hofmann, & Hayes, 2022; Hayes & Hofmann, 2021; Hofmann & Asmundson, 2008; Zettle & Gird, 2017). It has been proposed that MABIs incorporate a set of interrelated practices to cultivate a present-moment awareness and mindful attention (Hayes, Strosahl, & Wilson, 2012; Hofmann & Asmundson, 2008), train metacognitive skills such as decentering and distancing (Bernstein et al., 2015), with aim to facilitate an accepting and compassionate stance toward one's experience rather than attempting to avoid or otherwise control it (Johannsen, Nissen, Lundorff, & O'Toole, 2022; Hayes, Villatte, Levin, & Hildebrandt, 2011; van der Velden et al., 2015).

In this regard, several cognitive variables have been proposed to associate with the positive effects of MABIs. Among them, metacognitions (Jankowski & Holas, 2014; Norman et al., 2019; Sanger & Dorjee, 2016; Satlof-Bedrick & Johnson, 2015) and executive functions (Johannsen et al., 2022; Hölzel et al., 2011; Im et al., 2021; Swainston & Derakhshan, 2021) have received considerable theoretical and research attention and have been suggested as cognitive variables through which MABIs may exert their positive effects.

Metacognition refers to the awareness and understanding of one's own cognitive processes and broadly serves at least two basic functions: monitoring and control of cognition (Flavell & Wellman, 1977; Nelson & Narens, 1994). The monitoring function of metacognition involves being aware of and observing one's cognitive processes. On the other hand, the control function of metacognition involves using the information gathered through monitoring to regulate and direct cognitive processes (Efklides, 2011; Jankowski & Holas, 2014; Norman et al., 2019). Building on these two basic functions, cognitive scientists have attempted to categorize metacognitive phenomena. A classic distinction (Efklides, 2008, 2011; Flavell & Wellman, 1977) is between three facets of metacognition: knowledge, experiences, and strategies. Metacognitive knowledge involves understanding what one knows about their cognitive processes, such as their strengths, weaknesses, and higher order thinking skills (see for example Jankowski & Holas, 2014; Norman et al., 2019). Metacognitive experiences encompass the awareness of one's cognitive states and emotions during a cognitive task, providing insights into the affective aspects of thinking (Efklides, 2006; Jankowski & Holas, 2014). Metacognitive strategies refer to the way in which one voluntary and deliberately engages in various activities based on executive functions to control cognitions. More specifically, metacognitive strategies include planning, monitoring, and evaluating one's cognitive processes and relate to executive functions operating in order to control processes on the lower level (Efklides, 2008).

Executive functions are a set of higher-order cognitive processes that regulate and control one's goal-directed thoughts, actions, and attention (Friedman & Miyake, 2017). They allow individuals to successfully break out of habits, make decisions and evaluate risks, plan for the future, prioritize and sequence our actions, and cope with novel situations (Nyongesa et al., 2019; Snyder, Miyake, & Hankin, 2015). Metacognitions and executive functions are interconnected and complementary processes in cognitive functioning, as metacognitions often rely on executive functions for planning, monitoring, and controlling cognitive processes (Norman et al., 2019).

Accumulating clinical and experimental neuroscience evidence reveals that deficits in metacognitions and executive functions can act as predisposing risk factors, maintaining components, or negative consequences of anxiety disorders (Berggren & Derakshan, 2013; Crocker et al., 2013; Demetriou et al., 2021; LeMoult & Gotlib, 2019; Roskes, Elliot, Nijstad, & De Dreu, 2013; Sharp, Miller, & Heller, 2015; Snyder et al., 2015; Warren, Heller, & Miller, 2021). Maladaptive metacognitive beliefs, such as the belief that worrying is productive, may contribute to the maintenance of anxiety symptoms (McEvoy, 2019; Wells, 2000). Moreover, executive dysfunctions play a central role in anxiety disorders as they are often characterized by attentional biases towards threats and repetitive negative thinking patterns (Majeed et al., 2023; Zainal & Newman, 2018). Notably, findings show that impairments in these cognitive processes often persist in phases of symptom improvement and during remission (Levens & Gotlib, 2015; Schmid & Hammar, 2013). These findings imply that these processes hold particular promise for advancing science regarding the etiology, maintenance and treatment of anxiety disorders (Murphy et al., 2018). Moreover, preliminary studies suggest that MABIs improve metacognitive beliefs (Sanger & Dorjee, 2016; Zemestani and Ottaviani, 2016) and executive function capacities such as sustained attention (Morrison, Goolsarran, Rogers, & Jha, 2014) and decreased mind wandering (Mrazek, Smallwood, & Schooler, 2012).

Although MABIs have Eastern philosophical roots, much of the scientific literature regarding their effects on cognitive and metacognitive factors arises from North America (Brazier, 2013; Cuijpers et al., 2019; Trammel, 2017) and there is a significant paucity of scientific literature in this field among non-Western populations. The present study aimed to address this gap and providing cross-cultural evidence in support of MABIs as interventions that potentially target cognitive and metacognitive factors associated with anxiety disorders in a non-Western culture. In designing MABI for the present study, we focused on the integration of major Iranian culture. Because MABI and many aspects of Islam are compatible, it was important to build on the connection between MABI and Islamic culture in Iran. Our hope was that the incorporation of MABI and some spiritual Islamic concepts would ensure easier assimilation of program concepts into the lives of the participants. We hypothesized that MABI would show superior efficacy in comparison to WLC for decreasing scores on anxiety symptoms as primary outcome as well as decreasing scores on dysfunctional metacognitive beliefs, executive dysfunctions, and increasing scores on mindfulness facets as secondary outcomes.

2. Methods

2.1. Participants and procedure

Participants were forty-five university students with anxiety disorders (mean [SD] age, 24.60 [4.94] years; 51.1 % female), recruited from University of Kurdistan (Sanandaj, Iran). The required sample size was calculated a priori based on a medium effect size using the G*Power program and resulted in a minimum total sample size of 28 (repeated measures ANOVA, mixed-factorial design, $\alpha = 0.05$, power $= 0.80$). Inclusion criteria were (a) scoring moderate or higher on the Beck Anxiety Inventory (BAI total score >16); (b) meeting DSM-5 criteria for at least one current anxiety disorder as assessed by a trained master-level psychologist using the Persian version of the SCID-5-CV (First, Williams, Karg, & Spitzer, 2016; Mohammadkhani, Forouzan, Hooshyari, & Abasi, 2020), (c) being a university student, (d) being between 18 and 65 years, (e) being able to speak fluent in Persian language as a formal and commonly spoken language in Iran universities, (f) and being willing to provide voluntary informed consent. Exclusion criteria were

(a) having acute psychiatric conditions including psychotic disorder, bipolar disorder, substance abuse disorder, and suicidal ideation, (b) and receiving psychotropic medication or psychotherapy during the last six months.

Despite the predominance of female students in Iranian universities, gender norms, cultural factors, and mental health stigma are reported as potential barriers to their enrollment in mental health programs (Sarikhani et al., 2021; Zemestani et al., 2024; Zemestani, Hosseini et al., 2022). We enacted gender-matched recruitment to achieve a more balanced gender distribution and enhance the external validity of study (Daitch et al., 2022). Moreover, we utilized online platforms and social media channels to reach a wider audience and facilitate participation and address the unique challenges posed by gender norms and mental health stigma. Participants were recruited through announcements in the university social media channels and groups including WhatsApp, Viber, Telegram or Instagram. Interested participants first were asked to review and complete an online informed consent form and then were directed to complete an online screening assessment using the BAI administered on the Google Form platform.

Among the students who completed the online screening assessment,

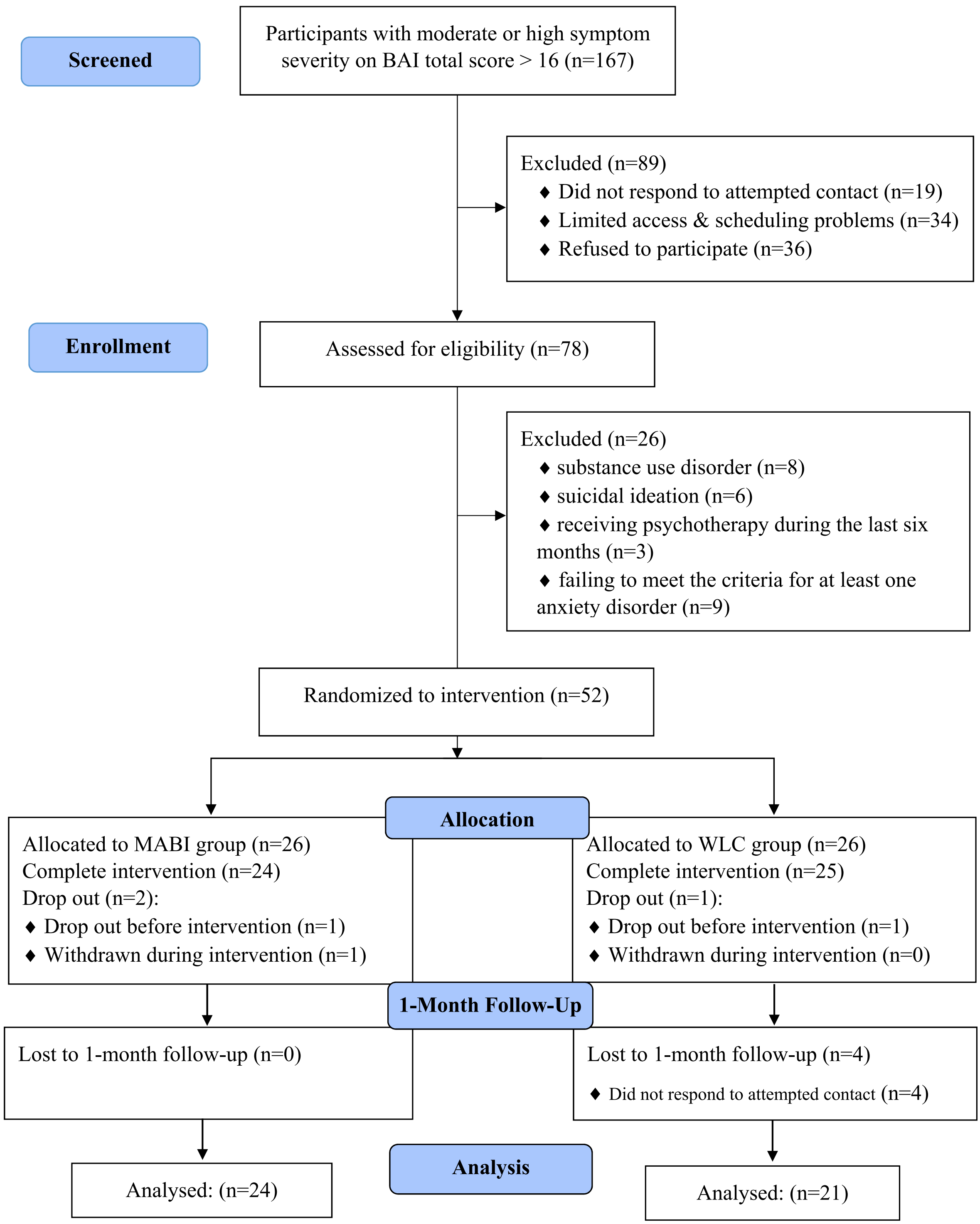

Fig. 1. CONSORT diagram for the process of participants' recruitment.

167 students had moderate or higher symptom severity on the BAI (total score >16). Among students with moderate or high symptom severity, 89 students were excluded during the eligibility assessment for the following reasons: did not respond to attempted contact (n = 19), limited access and scheduling problems (n = 34), or declined to participate (n = 36). The remaining 78 interested students were invited to attend an interview session to assess study eligibility at the University of Kurdistan counseling center. Interviews were conducted by a trained masters-level psychologist using the Persian version of the SCID-5-CV (First et al., 2016; Mohammadkhani et al., 2020). After eligibility interviews, 26 students did not meet the inclusion criteria for the following reasons: substance use disorder (n = 8), suicidal ideation (n = 6), receiving psychotherapy during the last six months (n = 3), or failing to meet the criteria for at least one anxiety disorder (n = 9). A total of 52 eligible and interested participants who met the inclusion criteria and agreed to participate in the study were randomized to either the MABI (n = 26) or WLC (n = 26) groups. Randomization, in ratio of 1:1, was conducted using a computer-generated algorithm in a parallel study design to ensure balanced groups. Only participants who completed the trial were included in the analysis (MABI = 24, WLC = 21). The CONSORT diagram is presented in Fig. 1.

2.2. Ethics

Study was approved by the University of Kurdistan Research Ethics Committee [coded as: IR.UOK.REC.1398.045], which followed the ethical standards outlined in the Declaration of Helsinki regarding research on human participants. Participants consented to intervention, and written informed consent was obtained from all participants in the research study.

2.3. Mindfulness and acceptance-based intervention (MABI)

The MABI consisted of nine in-person weekly 60-min group sessions of mindfulness and acceptance-based practices following the guidelines laid out by Hayes et al. (2012), and Kabat-Zinn (2013). The mindfulness program included a stronger emphasis on brief formal mindfulness and meditation practices. Formal practices included body scan meditation, mindful yoga, sitting meditation with focus on the breath, sitting meditation expanding awareness to objects of attention beyond the breath to open awareness of the present moment, walking meditation, and eating meditation. Informal mindfulness practices included mindfulness of everyday activities such as mindful eating, walking, watching, showering, and so forth. Class discussions focused on group members' experiences in the formal meditation practices and the application of informal mindfulness in day-to-day life.

Each session had a central theme and included didactic presentations and formal mindfulness practices aimed to improve cognitive flexibility, metacognitive beliefs, executive functions and reduce anxiety symptoms. Sessions involved a variety of experiential exercises, interspersed with discussions of the role of mindfulness and acceptance in psychological flexibility and metacognitive capacities. The overarching theme of momentary awareness and acceptance of negative thoughts and emotions related to stress and anxiety was introduced and reinforced in complementary ways throughout the training. Approximately 30 min of daily home practice of formal and informal mindfulness practices was assigned and encouraged between sessions.

While the protocol adaptation was not part of the formal study process, in designing MABI for the present study we also focused on the integration of major Iranian culture. Because MABI and many spiritual aspects of Islam are compatible, it was important to build on the connection between MABI and Islamic culture in Iran. Our hope was that the incorporation of MABI and some spiritual concepts would ensure easier assimilation of program concepts into the lives of the participants. Adaptations to the treatment manual were based on linguistic and cultural adaptations of MABIs for Iranian individuals with anxiety disorder symptoms as well as the clinical experiences of the study team in treating Iranian individuals with anxiety disorders. For example, linguistic adaptations were used for the participants' cultural syndromes and the participants' idioms of distress and anxiety (e.g., concerns about "distress" or "*parishani*," and "anxiety" or "*ezterab*"). As another example, the self-as-context metaphor of the "chessboard" was adapted into the "*shatranj*" metaphor. Shatranj is the Farsi term for modern-day chess, stemming from its ancient origins, and is well-known in Iranian culture. Additionally, all guided exercises were translated and recorded in Persian (e.g., leaves on a stream). Please see appendix for a summary of MABI session content.

The sessions were run by a trained Masters-level cognitive psychologist with at least one-year experience. Prior to the study, the therapist received supervised training in delivering mindfulness- and acceptance-based interventions for anxiety disorders and received certification in relevant specialized workshops. The supervisor was an Associate Professor in psychology specializing in cognitive-behavioral therapy as well as MABIs and was not directly involved in conducting the intervention.

2.4. Waitlist control (WLC)

Participants in the WLC condition were asked to come into the counseling center to review their mental health status and complete study assessments in-person at baseline, 9-weeks (equivalent to post-intervention), and 13-weeks (equivalent to one-month follow-up), in 45-min duration sessions. In order to comply with research ethics, following completion of the WLC, participants were offered nine group sessions of MABI.

2.5. Measures

2.5.1. Structured clinical interview for DSM-5 disorders-clinician version (SCID-5-CV)

The SCID-5-CV (First et al., 2016) is a semi-structured interview guide for making the major DSM-5 diagnoses. In this study, we used the SCID-5-CV to determine study eligibility and to assess anxiety disorders, depressive disorders, substance use disorders, bipolar, and psychotic disorders. The SCID-5-CV has demonstrated good internal consistency (a >0.80), as well as excellent reliability and validity in the diagnosis of severity of mental disorders (Shankman et al., 2018). We used the Persian version of the SCID-5-CV (Mohammadkhani et al., 2020), which has demonstrated good psychometric properties in Iranian population.

2.5.2. Beck Anxiety Inventory (Bai)

The BAI (Beck & Steer, 1990) is a widely used 21-item self-report inventory assessing anxiety symptoms. Each item is rated on a four-point Likert scale ranging from 0 (not at all) to 3 (severely) with a total score ranging from 0 to 63. The BAI total score of 0–7 indicates minimal anxiety, 8–15 mild anxiety, 16–25 moderate anxiety, and 26–63 severe anxiety. The BAI possesses good psychometric properties with Cronbach's $\alpha = 0.92$ (Beck & Steer, 1990). We used the Persian version of the BAI (Kaviani & Mousavi, 2008), which has demonstrated excellent internal consistency ($\alpha = 0.92$), and good test-retest reliability ($r = 0.72$). Internal consistency reliability in the current study was $\alpha = 0.71$.

2.5.3. Metacognitions Questionnaire-30 (MCQ-30)

The MCQ-30 (Wells & Cartwright-Hatton, 2004) is a 30-item self-report measure designed to assess five dimensions of metacognitions: positive metacognitive beliefs (e.g., "Worrying helps me cope"), negative metacognitive beliefs (e.g., "When I start worrying, I cannot stop"), cognitive confidence (e.g., "I have a poor memory"), cognitive self-consciousness (e.g., "I pay close attention to the way my mind works"), and need to control thoughts (e.g., "Not being able to control my thoughts is a sign of weakness"). Each item is rated on a 4-point Likert scale (1 = Do not agree to 4 = Agree too much). A total score can be calculated, with higher scores indicate higher levels of maladaptive and dysfunctional metacognitions. Cronbach's α for the total score and five subscales reported in the range between 0.72 and 0.93, demonstrating good–excellent internal consistency (Wells & Cartwright-Hatton, 2004). We used the Persian version of the MCQ-30 (Shirinzadeh, Goudarzi, & Naziri, 2009), which has demonstrated excellent internal consistency for the total score (α = 0.91). Internal consistency reliability for the total score in the current study was α = 0.74.

2.5.4. Behavior Rating Inventory of Executive Function-Adult Version (BRIEF-A)

The BRIEF-A (Roth, Gioia, & Isquith, 2005) is a 75-item self-report measure designed to assess perception of behavioral and emotional manifestations of executive dysfunction in daily life. This measure yields an overall score of executive function (Global Executive Composite; GEC) and two indices of Behavioral Regulation Index (BRI), composed of 4 scales (i.e., inhibit, shift, emotional control, and self-monitor), and Metacognition Index (MI), composed of 5 scales (i.e., initiate, working memory, plan/organize, task monitor, and organization of materials). Each item is rated on three response choices ("never," "sometimes," or "often"), with higher scores indicating poorer executive functions. In the normative sample (Roth et al., 2005), moderate to high internal consistency has been reported for the scales (α = 0.73 to 0.90), as well as for the indices and the GEC (α = 0.93 to 0.96) and 1-month test–retest reliabilities (ranging from r = 0.93 to 0.94) for the three major indices. We used the Persian version of the BRIEF-A (Mohammadnia, Bigdeli, Mashhadi, Ghanaei Chamanabad, & Roth, 2022), which has demonstrated excellent internal consistency for the total score (α = 0.93). In the current study, internal consistency reliability for the total score was α = 0.78.

2.5.5. Five Facet Mindfulness Questionnaire (FFMQ)

The FFMQ (Baer, Smith, & Allen, 2004) is composed of 39 items distributed on five scales assessing mindfulness content, including observation, description, aware activity, non-judgment, and non-reactivity. Participants were asked to rate their level of agreement with the phrases, on a 1–5-point Likert scale (1 = "never or almost never", 5 = "always or almost always"). The internal reliabilities of the scales range between 0.81 and 0.87, and the correlations between them are between 0.31 and 0.67 (Baer et al., 2004). We used the Persian version of the FFMQ (Ahmadvand, Heydarinasab, & Shaeiri, 2013), which has demonstrated excellent internal consistency for the total score (α = 0.80). In the present study, internal consistency reliability for the total scores of scale was α = 0.74.

2.6. Statistical analysis

IBM Statistical Package for the Social Sciences (SPSS Inc, New York, USA) Version 27 was used to analyze the data. Participants demographic and baseline characteristics were compared using the chi-square test for categorical variables and independent two-samples *t*-test for continuous variables. To evaluate intervention effect on outcomes, repeated measure ANOVAs were performed, entering the primary outcome (BAI) and process of change measures (MCQ-30, BRIEF-A, and FFMQ) as dependent variables, group (MABI vs. WLC) as the between-subject factor, and time (pre-, post-, follow-up) as the within-subject factor. Kolmogorov-Smirnov, Levin's and Mauchly's tests were used to evaluate the normality, homogeneity of variance and sphericity of the data, respectively, before performing the repeated measures ANOVA. Within- and between-group effect sizes were reported based on Cohen's *d* using the SPSS-27. Post hoc analyses were conducted using exploratory *Studentś* t-tests and were not corrected for multiple pairwise comparisons, given the small sample size (two-tailed, critical $p < 0.05$).

3. Results

Table 1 provides a summary of the sociodemographic clinical characteristics of the study participants. Forty-five participants completed the trial (MABI = 24, $M_{age} = 25.46$, $SD = 6.17$; WLC = 21; $M_{age} = 23.62$, $SD = 2.89$), indicating overall attrition was low. The total sample included 22 males and 23 females, with a mean age of 24.60 years (SD = 4.95), ranging 18–45 years. All participants had a comorbid anxiety disorder diagnosis. There were no significant differences between groups on demographic and clinical characteristics of participants and outcome variables at pretreatment ($p > 0.05$). Data were normally distributed, and homogeneity of variance and sphericity of the data were confirmed by the Kolmogorov-Smirnov, Levin's and Mauchly's tests, respectively ($p < 0.05$). Means, standard deviations, and within-group effect sizes of outcomes are shown in Table 2.

3.1. The effects of MABI on primary outcome

A repeated measure ANOVA tested for an overall time by group effect on the primary outcome measures of emotional disorders symptoms between two groups (MABI vs. WLC) and over three time points (pre, post, follow-up). As to primary outcome measures (BAI), a significant interaction effect of time × group was observed (BAI; $F_{(2, 42)} = 46.14$, $p < 00.001$, Cohen $d = -1.89$, 95% CI [2.60 ± 1.18]), in favor of MABI over WLC; see Table 3). Post hoc between-group comparisons using exploratory *Students* t-test showed significantly diminished anxiety symptoms at post-intervention and follow-up measurements (*ps* < 0.001) in the MABI group but not the WLC group. Within-group

Table 1
Demographic and clinical characteristics of the study participants.

Variable	MABI (n = 24)	WLC (n = 21)	Total (N = 45)	t or χ^2	P [a]
Age: mean (±SD)	25.46 (6.17)	23.62 (2.89)	24.60 (4.95)	−1.252	0.217
Gender: *n* (%)				0.192	0.461
Male	11 (45.8%)	11 (52.4%)	22 (48.9%)		
Female	13 (54.2%)	10 (47.6%)	23 (51.1%)		
Education: *n* (%)				5.228	0.156
Associate degree	3 (12.5%)	1 (4.8%)	4 (8.9%)		
Bachelor degree	9 (37.5%)	15 (71.4%)	24 (53.3%)		
Master degree	10 (41.7%)	4 (19.04%)	14 (31.1%)		
Ph.D. degree	2 (8.3%)	1 (4.8%)	3 (6.7%)		
Marriage: *n* (%)				1.091	0.296
Single	19 (79.2)	19 (90.5%)	38 (84.4%)		
Married	5 (20.8)	2 (9.5%)	7 (15.6%)		
Anxiety disorders: *n* (%)				24.911	<0.001
Generalized anxiety disorder	7 (29.2%)	8 (38.1%)	15 (33.3%)		
Social anxiety disorder	4 (16.7%)	3 (14.3%)	7 (15.5%)		
Specific phobia	3 (12.5%)	3 (14.3%)	6 (13.3%)		
Agoraphobia	3 (12.5%)	2 (9.5%)	5 (11.1%)		
Panic disorder	1 (4.2%)	1 (4.7%)	2 (4.4%)		
Separation anxiety	4 (16.7%)	2 (9.5%)	6 (13.3%)		
Other specified anxiety disorder	2 (8.3%)	2 (9.5%)	4 (8.9%)		

Note. MABI = Mindfulness and Acceptance-Based Intervention; WLC= Wait-List Control.
[a] Based on *t*-test for age and χ^2-test for all other demographic variables.

Table 2
Symptoms and functioning over time: mean, standard deviations, and within-condition effect sizes.

Measure	MABI (n = 24)		WLC (n = 21)	
	Mean (SD)	Cohen's D [95% CI]	Mean (SD)	Cohen's D [95% CI]
BAI				
Pre-treatment [a]	29.33 (7.05)		29.76 (6.89)	
Post-treatment [b]	18.79 (3.67)	1.69 [1.05 ± 2.31]	29.19 (6.99)	0.46 [0.02 ± 0.90]
Follow-up [c]	18.04 (3.72)	1.59 [0.94 ± 2.17]	29.24 (7.04)	0.53 [0.07 ± 0.98]
MCQ-Total				
Pre-treatment	71.45 (6.74)		70.76 (5.32)	
Post-treatment	58.20 (4.45)	1.71 [1.07 ± 2.34]	71.09 (4.28)	−0.09 [-0.52 ± 0.33]
Follow-up	58.16 (4.94)	1.19 [1.22 ± 2.58]	71.19 (4.65)	−0.11 [-0.54 ± 0.31]
MCQ-PB				
Pre-treatment	12.95 (2.38)		13.09 (3.14)	
Post-treatment	11.12 (2.17)	0.70 (0.24 ± 1.14)	13.42 (2.97)	−0.21 (−0.64 ± 0.22)
Follow-up	10.62 (1.88)	0.76 (0.30 ± 1.12)	13.33 (2.72)	−0.10 (−0.53 ± 0.32)
MCQ-NB				
Pre-treatment	14.87 (2.68)		16.57 (2.35)	
Post-treatment	11.58 (2.74)	0.72 (0.26 ± 1.12)	16.19 (2.08)	0.29 (−0.14 ± 0.73)
Follow-up	12.29 (2.58)	0.70 (0.26 ± 1.14)	16.14 (1.90)	0.31 (−0.12 ± 0.74)
MCQ-CC				
Pre-treatment	13.91 (1.95)		11.09 (1.37)	
Post-treatment	11.16 (2.03)	1.02 (0.52 ± 1.51)	11.42 (1.39)	−0.28 (−0.72 ± 0.15)
Follow-up	11.54 (1.21)	0.85 (0.31 ± 1.31)	11.38 (1.46)	−0.24 (−0.67 ± 0.19)
MCQ-CS				
Pre-treatment	13.50 (2.62)		14.57 (2.71)	
Post-treatment	11.41 (2.14)	0.69 (0.24 ± 1.13)	14.61 (2.59)	0.02 (−0.45 ± 0.40)
Follow-up	11.04 (1.65)	1.25 (0.71 ± 1.78)	15.28 (3.11)	0.25 (−0.64 ± 0.18)
MCQ-NC				
Pre-treatment	16.20 (2.88)		15.42 (2.92)	
Post-treatment	12.91 (3.04)	0.70 (0.25 ± 1.15)	14.42 (2.73)	0.01 (−0.42 ± 0.42)
Follow-up	12.66 (2.33)	1.01 (0.51 ± 1.49)	14.04 (2.97)	0.18 (−0.24 ± 0.61)
BRIEF-A				
Pre-treatment	79.67 (5.30)		79.43 (7.63)	
Post-treatment	68.67 (6.32)	2.75 [1.86 ± 3.63]	79.24 (7.79)	0.06 [-0.37 ± 0.48]
Follow-up	68.54 (6.06)	2.70 [1.82 ± 3.55]	79.38 (7.65)	−0.13 [-0.56 ± 0.30]
FFMQ				
Pre-treatment	109.46 (15.64)		110.33 (14.14)	
Post-treatment	133.63 (9.46)	−1.81 [-2.45 ±-1.43]	110.52 (14.28)	−0.10 [-0.53 ± 0.32]
Follow-up	132.71 (9.17)	−1.86 [-2.52 ±-1.90]	110.57 (14.37)	−0.10 [-0.52 ± 0.32]

Note. MABI = Mindfulness and Acceptance-Based Intervention; WLC= Wait-List Control; BAI= Beck Anxiety Inventory; MCQ-Total = Metacognitive Questionnaire-Total scores; MCQ-PB= Positive Beliefs about Worry; MCQ-NB= Negative Beliefs about Worry; MCQ-CC= Cognitive Confidence; MCQ-CS= Cognitive Self-Consciousness; MCQ-NC= Need to Control Thoughts; BRIEF-A = Behavior Rating Inventory of Executive Function-Adult Version; FFMQ= Five Facet Mindfulness Questionnaire.

[a] Pre-treatment refers to the baseline appointment.

[b] Post-treatment or 12-weeks for WLC condition.

[c] One-month post-treatment follow-up or 16-weeks for WLC condition.

Table 3
Results of the repeated measure ANOVA on outcome measures.

Measure	Group	Time	Group × Time	Cohen's D [a] [95% CI]
BAI	$F_{(1, 43)} = 19.68$ $P < 0.001$	$F_{(2, 42)} = 57.53$ $P < 0.001$	$F_{(2, 42)} = 46.14$ $P < 0.001$	−1.89 [2.60 ± 1.18]
MCQ-Total	$F_{(1, 43)} = 45.88$ $P < 0.001$	$F_{(2, 42)} = 44.08$ $P < 0.001$	$F_{(2, 42)} = 49.45$ $P < 0.001$	2.94 [-3.79 ±-2.08]
MCQ-PB	$F_{(1, 43)} = 6.63$ $P < 0.013$	$F_{(2, 42)} = 5.44$ $P < 0.006$	$F_{(2, 42)} = 8.92$ $P < 0.001$	−0.89 [-1.50 ±-0.27]
MCQ-NB	$F_{(1, 43)} = 23.09$ $P < 0.001$	$F_{(2, 42)} = 10.56$ $P < 0.001$	$F_{(2, 42)} = 6.28$ $P < 0.003$	−1.87 [2.56 ±-1.15]
MCQ-CC	$F_{(1, 43)} = 4.36$ $P < 0.043$	$F_{(2, 42)} = 10.06$ $P < 0.001$	$F_{(2, 42)} = 16.35$ $P < 0.001$	−0.17 [-0.75 ±-0.41]
MCQ-CS	$F_{(1, 43)} = 20.72$ $P < 0.001$	$F_{(2, 42)} = 4.84$ $P < 0.010$	$F_{(2, 42)} = 10.48$ $P < 0.001$	−1.35 [-1.99 ±-0.69]
MCQ-NC	$F_{(1, 43)} = 4.26$ $P < 0.045$	$F_{(2, 42)} = 10.90$ $P < 0.001$	$F_{(2, 42)} = 8.54$ $P < 0.001$	−0.86 [-1.47 ±-0.25]
BRIEF-A	$F_{(1, 43)} = 13.07$ $P < 0.001$	$F_{(2, 42)} = 87.52$ $P < 0.001$	$F_{(2, 42)} = 83.86$ $P < 0.001$	−1.50 [-2.15 ±-0.83]
FFMQ	$F_{(1, 43)} = 16.66$ $P < 0.001$	$F_{(2, 42)} = 65.55$ $P < 0.001$	$F_{(2, 42)} = 63.24$ $P < 0.001$	1.93 [1.21 ± 2.64]

Note. BAI= Beck Anxiety Inventory; MCQ-Total = Metacognitive Questionnaire-Total scores; MCQ-PB= Positive Beliefs about Worry; MCQ-NB= Negative Beliefs about Worry; MCQ-CC= Cognitive Confidence; MCQ-CS= Cognitive Self-Consciousness; MCQ-NC= Need to Control Thoughts; BRIEF-A = Behavior Rating Inventory of Executive Function-Adult Version; FFMQ= Five Facet Mindfulness Questionnaire.

[a] Between-condition effect sizes (Group × Time interaction.

comparisons showed a significant decrease of anxiety symptoms in the MABI, but not the WLC group from pre-to post-intervention and pre-intervention to follow-up (see Table 4).

3.2. The effects of MABI on process of change variables

Another set of repeated measure ANOVA tested for time × group effects on the process of change variables underlying anxiety disorders. Results indicated a significant interaction effect of time × group for total and subscales of metacognitions (MCQ-Total; $F_{(2, 42)} = 49.45$, $p < 00.001$, Cohen $d = 2.94$, 95% CI [−3.79±-2.08]; MCQ-PB; $F_{(2, 42)} = 8.92$, $p < 00.001$, Cohen $d = -0.89$, 95% CI [−1.50±-0.27]; MCQ-NB; $F_{(2, 42)} = 6.28$, $p < 00.003$, Cohen $d = -1.87$, 95% CI [2.56±-1.15]; MCQ-CC; $F_{(2, 42)} = 16.35$, $p < 00.001$, Cohen $d = -0.17$, 95% CI [−0.75 ±-0.41]; MCQ-CS; $F_{(2, 42)} = 10.48$, $p < 00.001$, Cohen $d = -1.35$, 95% CI [−1.99±-0.69]; MCQ-NC; $F_{(2, 42)} = 8.54$, $p < 00.001$, Cohen $d = -0.86$, 95% CI [−1.47±-0.25]) and executive functions (BRIEF-A; $F_{(2, 42)} = 83.86$, $p < 00.001$, Cohen $d = -1.50$, 95% CI [−2.15±-0.83]), as well as mindfulness facets (FFMQ; $F_{(2, 42)} = 63.24$, $p < 00.001$, Cohen $d = 1.93$, 95% CI [1.21 ± 2.64]), in favor of MABI over WLC (see Table 3). Post hoc between-group comparisons showed significant gains in metacognitive beliefs and executive functions at post-intervention and follow-up measurements (*ps* < 0.001) in the MABI group but not the WLC group. Within-group comparisons showed a significant change in metacognitive beliefs and executive functions in the MABI, but not the WLC group from pre-to post-intervention and pre-intervention to follow-up (see Table 4).

Table 4
Results of the post hoc exploratory *t*-test for pairwise comparisons (within- and between-group changes).

Measure	Time	MABI (n = 24)			WLC (n = 21)		
		Mean differences (t) [a]	*P*	95% CI	Mean differences (t) [a]	*P*	95% CI
BAI	Pre to Post	10.542	<0.001	8.180 ± 12.903	0.571	1.000	−1.953 ± 3.096
	Pre to FU	11.292	<0.001	8.605 ± 13.978	0.524	1.000	−2.348 ± 3.396
	Post to FU	0.750	0.029	0.59 ± 1.441	−0.048	1.000	−0.787 ± 0.691
	Pre^{MBSR} vs Pre^{WLC}	0.429	0.838	−3.776 ± 4.633	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	10.399	<0.001	7.099 ± 13.698	–	–	–
	FU^{MBSR} vs FU^{WLC}	11.196	<0.001	7.872 ± 14.521	–	–	–
MCQ-Total	Pre to Post	13.25	<0.001	9.99 ± 16.50	−0.33	0.677	−1.97 ± 1.31
	Pre to FU	12.29	<0.001	10.35 ± 16.22	−0.42	0.607	−2.14 ± 1.28
	Post to FU	0.041	0.963	−1.81 ± 1.89	−0.09	0.880	−1.38 ± 1.19
	Pre^{MBSR} vs Pre^{WLC}	−0.13	0.869	−1.80 ± 1.52	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−2.30	<0.01	−3.85 ± −0.45	–	–	–
	FU^{MBSR} vs FU^{WLC}	−2.70	<0.001	−4.10 ± −1.31	–	–	–
MCQ-PB	Pre to Post	1.83	<0.01	0.72 ± 2.93	−0.33	0.339	−1.04 ± 0.37
	Pre to FU	2.33	<0.001	1.04 ± 3.62	−0.23	0.624	−1.23 ± 0.75
	Post to FU	0.50	0.963	−0.12 ± 1.12	0.09	0.789	−0.63 ± 0.82
	Pre^{MBSR} vs Pre^{WLC}	−1.69	0.118	−3.84 ± 0.450	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−4.60	<0.001	−6.09 ± −3.12	–	–	–
	FU^{MBSR} vs FU^{WLC}	−3.85	<0.001	−5.33 ± −2.37	–	–	–
MCQ-NB	Pre to Post	3.29	<0.01	1.35 ± 5.21	0.38	0.189	−0.20 ± 0.96
	Pre to FU	2.58	<0.01	0.82 ± 4.33	0.428	0.165	−0.19 ± 1.04
	Post to FU	−0.70	0.105	−1.57 ± 0.15	0.472	0.666	−0.17 ± 0.27
	Pre^{MBSR} vs Pre^{WLC}	−1.69	0.11	−3.84 ± 0.45	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−4.60	<0.001	−6.09 ± −3.12	–	–	–
	FU^{MBSR} vs FU^{WLC}	−3.85	<0.001	−5.33 ± −2.37	–	–	–
MCQ-CC	Pre to Post	2.75	<0.001	1.62 ± 3.87	−0.33	0.201	−0.85 ± 0.19
	Pre to FU	2.37	<0.001	1.19 ± 3.2502	−0.28	0.284	−0.82 ± 0.25
	Post to FU	−0.37	0.356	−1.19 ± 0.44	0.04	0.715	−0.22 ± 0.31
	Pre^{MBSR} vs Pre^{WLC}	−1.83	0.346	5.71 ± 0.93	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−2.13	<0.001	3.78 ± 0.62	–	–	–
	FU^{MBSR} vs FU^{WLC}	−3.56	<0.001	−4.12 ± 0.12	–	–	–
MCQ-CS	Pre to Post	−1.69	0.118	−3.84 ± 0.450	−0.04	0.917	−0.98 ± 0.89
	Pre to FU	2.45	<0.001	1.63 ± 3.28	−0.71	0.261	−2.02 ± 0.57
	Post to FU	0.375	0.377	−0.48 ± 1.23	−0.66	0.158	−1.61 ± 0.28
	Pre^{MBSR} vs Pre^{WLC}	−1.07	0.185	−2.67 ± 0.53	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−3.20	<0.001	4.62 ± −1.77	–	–	–
	FU^{MBSR} vs FU^{WLC}	−4.24	<0.001	5.71 ± −2.77	–	–	–
MCQ-NC	Pre to Post	3.29	<0.01	1.32 ± 5.25	0.01	1.000	−0.86 ± 0.85
	Pre to FU	3.54	<0.001	2.06 ± 5.02	0.38	0.40	−0.55 ± 1.31
	Post to FU	0.250	0.669	−0.94 ± 1.44	0.38	0.30	−0.37 ± 1.13
	Pre^{MBSR} vs Pre^{WLC}	0.77	0.374	−0.97 ± 2.53	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−2.51	<0.01	−4.26 ± −0.76	–	–	–
	FU^{MBSR} vs FU^{WLC}	−2.38	<0.01	−3.97 ± −0.78	–	–	–
BRIEF-A	Pre to Post	11.000	<0.001	9.134 ± 12.866	0.190	1.000	−1.805 ± 2.186
	Pre to FU	11.125	<0.001	9.287 ± 12.963	0.048	1.000	−1.917 ± 2.012
	Post to FU	0.125	1.000	−1.064 ± 1.314	−0.143	1.000	−1.414 ± 1.125
	Pre^{MBSR} vs Pre^{WLC}	−0.238	0.903	−4.150 ± −3.674	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	10.571	<0.001	6.352 ± 14.818	–	–	–
	FU^{MBSR} vs FU^{WLC}	10.839	<0.001	6.714 ± 14.964	–	–	–
FFMQ	Pre to Post	−24.167	<0.001	−29.184 ± −19.149	−0.190	1.000	−5.555 ± 5.174
	Pre to FU	−23.250	<0.001	−27.955 ± −18.549	−0.238	1.000	−5.268 ± 4.791
	Post to FU	0.917	0.649	−0.903 ± 2.737	−0.048	1.000	−1.993 ± 1.898
	Pre^{MBSR} vs Pre^{WLC}	0.875	0.846	−8.141 ± 9.891	–	–	–
	$Post^{MBSR}$ vs $Post^{WLC}$	−23.101	<0.001	−30.299 ± −15.903	–	–	–
	FU^{MBSR} vs FU^{WLC}	−22.137	<0.001	−29.293 ± −14.981	–	–	–

Note. MABI = Mindfulness and Acceptance-Based Intervention; WLC= Wait-List Control; BAI= Beck Anxiety Inventory; MCQ-Total = Metacognitive Questionnaire-Total scores; MCQ-PB= Positive Beliefs about Worry; MCQ-NB= Negative Beliefs about Worry; MCQ-CC= Cognitive Confidence; MCQ-CS= Cognitive Self-Consciousness; MCQ-NC= Need to Control Thoughts; BRIEF-A = Behavior Rating Inventory of Executive Function-Adult Version; FFMQ= Five Facet Mindfulness Questionnaire.

[a] Positive values denote a decrease between timeframes. $^{*}P < 0.05$.

4. Discussion

The aim of this study was to examine the effects of MABI on anxiety symptoms as well as metacognitions, executive functions, and mindfulness facets in Iranian individuals with anxiety disorders. Significant changes were observed over time on anxiety symptoms, as well as on metacognitive beliefs, executive dysfunctions, and mindfulness facets in favor of MABI over WLC. These results are in line with previous research findings regarding the effects of MABIs on metacognitions (Norman et al., 2019; Sanger & Dorjee, 2016) and executive dysfunction (Im et al., 2021; Johannsen et al., 2022; Swainston & Derakhshan, 2021). This RCT was conducted in a Middle East LMIC whereas much of the extant scientific literature about the effects of MABIs on cognitive and metacognitive factors associated with anxiety disorders arises from North America and Europe, and these factors are rarely examined in RCTs in non-Western countries (Brazier, 2013; Cuijpers et al., 2019; Trammel, 2017). Results from this RCT add to the knowledge in the field and provide cross-cultural evidence in support of MABIs as interventions that target anxiety symptoms, metacognitive beliefs and executive functions in a non-Western culture. Given the brief modular nature of the MABI and because of its potential to parsimoniously and flexibly address core symptoms of anxiety disorders and associated cognitive and metacognitive factors (i.e., executive functions and metacognitive beliefs), this intervention may be promising effective for the targeting cognitive and metacognitive factors associated with anxiety disorders among non-Western populations.

Theoretical accounts and clinical findings have proposed some cognitive mechanisms through which MABIs exert their positive effects and, among these, metacognitive processes (Jankowski & Holas, 2014; Norman et al., 2019; Sanger & Dorjee, 2016; Satlof-Bedrick & Johnson, 2015) and the executive function strategy of attention regulation (Johannsen et al., 2022; Hölzel et al., 2011; Im et al., 2021; Lyvers et al., 2014; Shapiro, Carlson, Astin, & Freedman, 2006; Swainston & Derakhshan, 2021) have been posited to promote sustained and selective attention and executive control of cognitive resources. It is possible that within the context of MABIs, metacognitive processes and executive functions work interactively to develop an observing stance towards one's thoughts, emotions, and bodily sensations. In general, MABI practices promote the ability to be aware of attentional focus in a given moment and to allocate cognitive resources in a goal-directed manner (Larson, Steffen, & Primosch, 2013; Veehof, Trompetter, Bohlmeijer, & Schreurs, 2016). Particularly, when MABIs are applied to anxiety disorders, one of the primary goals is to help individuals develop the ability to observe symptomatic processes without reacting to them in ways that exacerbate distress (Orsillo et al., 2005; Vøllestad, Nielsen, & Nielsen, 2012). By cultivating metacognitive awareness and enhancing executive function capacities, individuals are better equipped to navigate anxious experiences, regulate their attention, and respond to anxiety in a more adaptive and mindful manner (Jankowski & Holas, 2014). Through mindfulness practices, individuals learn to recognize and observe their thoughts without becoming fully absorbed or entangled in them (Sanger & Dorjee, 2016; Zemestani et al., 2016; Zemestani & Mozaffari, 2020).

Mindfulness induces a metacognitive mode of information processing that leads to beneficial changes in associated executive functions and self-regulation capacities which in turn reduces symptoms of anxiety and associated psychological distress. Metacognition ranges from knowledge about cognition and what could be regarded as higher order thinking skills (i.e., knowledge) and feeling states (i.e., experiences), to regulation of cognition (i.e., strategies). These three facets of metacognition (i.e., knowledge, experiences and strategies) are theoretically distinct, and successful cognitive and emotional processing relies on an efficient and goal-directed orchestration of each element, in interaction with each other (Efklides, 2008; Jankowski & Holas, 2014). The relation between meta-knowledge and meta-experiences is mutual: anxious feelings and emotions may trigger some elements of meta-knowledge, and meta-knowledge in turn can influence the way information from the object level are to be interpreted and experienced. Within the context of anxiety disorders, meta-knowledge and maladaptive metacognitive beliefs such as the belief that worrying is productive, may contribute to the maintenance of anxiety symptoms (McEvoy, 2019). Both meta-experiences and meta-knowledge impact meta-strategies or executive functions: the former play a motivational role and the later provides a proper script for the control processes (Efklides, 2008; Norman et al., 2019). Meta-strategies or executive functions play a central role in anxiety disorders as they are often characterized by attentional biases towards threats and repetitive negative thinking patterns (Majeed et al., 2023; Zainal & Newman, 2018). Notably, findings show that MABIs improve metacognitive beliefs (Sanger & Dorjee, 2016; Zemestani et al., 2016) and meta-strategies or executive function capacities (Morrison et al., 2014; Mrazek et al., 2012).

Previous research (e.g., Jankowski & Holas, 2014) suggested that MABI practices impose at least two levels of cognition: (1) the lower level which refers to the qualia (basic qualities of experience such as regulating attention, self-regulation and other executive function and metacognitive strategies) occurring in the present, and (2) the higher level constituted by awareness of the flowing qualia (higher-level of experience such as flaw states and other metacognitive awareness and metacognitive experiences). Important implication of putting mindfulness on the highest level of cognition is that mindful state is always conscious. In this model, the basic feature of mindfulness is meta-awareness – objective and conscious awareness of flowing internal events (Jankowski & Holas, 2014). Relating mindfulness to conscious, intentional regulation of attention implicates executive functions as an important element of the construct (Bishop et al., 2004; Holas & Jankowski, 2013; Shapiro et al., 2006). It means that person – when mindful – is aware of being aware, and is able to report not only a content of current consciousness but also the metacognitive knowledge and metacognitive experiences related to mindfulness. This suggests the cognitive control and other processes related to executive function often run unconsciously under the mindful states (Efklides, 2011; Jankowski & Holas, 2014).

On the other hand, the relative improvement of anxiety symptom severity, metacognitions, and executive function strategies following MABI may be explained by cognitive neuroscience perspective. Extensive research shows that engaging in meditation practice results in improved cognitive performance, which is associated with changes in brain activity indicative of improved neural efficiency (Chiesa, Calati, & Serretti, 2011; Lutz, Slagter, Dunne, & Davidson, 2008; Malinowski, 2013; Slagter, Davidson, & Lutz, 2011). More recently, cognitive scientists and neuroscientists, along with other researchers who study cognitive mechanisms underlying MABIs, have a considerable amount of work to make meaningful progress (Im et al., 2021; Van Dam et al., 2018). Studies in human neuroanatomical models of anxiety disorders, consistently identified the pathophysiological role of a 'fear network' comprising the prefrontal cortex (PFC), anterior cingulate cortex (ACC), and amygdala, as part of the extensive frontal-limbic network (Hölzel et al., 2007; Leicht & Mulert, 2020).

Experimental findings show the effects of mindfulness practices on executive functions, which control and direct cognitive processes for self-regulation, attention regulation, working memory, planning, decision making, and many other goal-directed behaviors (Black, Semple, Pokhrel, & Grenard, 2011; Gallant, 2016; Teper & Inzlicht, 2013), with particular reference to executive inhibition (Gallant, 2016; Moore & Malinowski, 2009). Such executive functions enhanced by mindfulness have been associated to the notion of cognitive flexibility (Kim, Johnson, Cilles, & Gold, 2011; Miller & Cohen, 2001; Moore & Malinowski, 2009), which is central construct in the third wave process-based interventions, particularly in the MABIs (Ciarrochi et al., 2022; Hayes & Hofmann, 2021). In general, the fact that MABIs improve both cognitive and metacognitive processes (i.e., executive functions and metacognitive beliefs) in anxious individuals suggests that neurocognitive processes may underlie the link between mindfulness and higher-level cognitions (Raffone & Srinivasan, 2017). Although an impact of mindfulness over the PFC and ACC on the outcome variables in our study cannot be ruled out on conceptual grounds, based on the pathophysiology of anxiety disorders with PFC and ACC changes, the intervention-induced PFC and ACC modification is most probably responsible for the observed effects (Lin, Fisher, & Moser, 2019).

The findings of present study add important implications about the impact of MABIs on core cognitive and metacognitive factors associated with anxiety disorders. Moreover, the current RCT was conducted in a LMIC Middle East country; as such, results provide a cross-cultural evidence in support of MABIs as interventions that target cognitive and metacognitive factors associated with anxiety disorders among a non-Western population. While further research on potential cognitive and metacognitive factors of MABIs is warranted, these findings are particularly encouraging and provide insights into designing more effective process-based therapeutic techniques for targeting cognitive and metacognitive factors associated with anxiety disorders.

The findings should be interpreted in the context of some limitations and suggestions for future research. The sample size of the study was small, and replication studies with larger sample sizes are required. The relatively small sample size did not permit assessment of effects for specific anxiety disorders and, instead, findings apply to a group with heterogeneous anxiety diagnoses. Future research on specific anxiety disorders is warranted. A self-report method was used to collect the data, which is prone to several biases (e.g., socio-cultural bias, recall bias, social desirability bias). Although internal consistency reliability for the BAI in the current study was comparable with its original Persian language psychometric study (Kaviani & Mousavi, 2008), acceptable internal consistencies was achieved for other self-report measures. Given the small sample size and relatively lower internal consistencies of the self-report measures, the large effect sizes reported in this study are especially robust. Future research should incorporate interview-based measures to reduce the potential role that social desirability bias and mental health stigma might play in outcomes. Post hoc t-tests analyses were conducted as exploratory and were not corrected for multiple pairwise comparisons, given the small sample size. Future research with larger sample size will include corrections for multiple pairwise comparisons, such as Bonferroni correction, to mitigate the risk of Type I error. In this study, we did not test the mechanistic factors involved in the intervention effect. Future studies should consider longitudinal mediation analysis to explore the underlying mechanisms of the impact of our mindfulness intervention on metacognitions, executive functions, and anxiety symptoms. Another limitation of this study is related to the study population. Iranian university students may not be fully representative of the broader Iranian population for several reasons including the age and gender distribution, educational level, and occupational status. Gender norms, sociocultural restrictions, and mental health stigma are reported as potential barriers to enrollment in mental health programs among Iranian individuals (Zemestani, Hosseini et al., 2022; Zemestani et al., 2024). All genders were in group MABI sessions together to facilitate participation and address the unique challenges posed by gender norms and mental health stigma. The low attrition and high retention rates in the study warrant explanation. Although we did not reimburse participants, which is uncommon, some motivational methods such as using regular reminders through mobile calls, text messages, and emails helped participants remember upcoming sessions and enhanced participant engagement and motivation to remain in study. Moreover, acknowledging treatment gains during the sessions and discussing how learned skills can be integrated into daily life may have helped reduce attrition. Moreover, MABI was compared against a WLC, rather than an active control condition with equal therapist contact and homework. Finally, the MABI sessions was administered by a Master's-level cognitive psychologist, calling into question whether the treatment might be effectively administered by providers with less formal training. As the time and costs required for training professional clinicians are two of the largest barriers to the dissemination of empirically supported treatments in LMICs (Cuijpers et al., 2019; Mascayano, Armijo, & Yang, 2015), this question warrants further exploration. Future research should use multiple Ph.D. level clinical psychologist to increase generalizability of findings.

CRediT authorship contribution statement

Reza Didehban: Project administration, Methodology, Formal analysis, Data curation. **Mehdi Zemestani:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Conceptualization. **Gordon J. G. Asmundson:** Writing – review & editing, Validation. **Jafar Bakhshaie:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

Appendix. Culturally modified MABI sessions for Iranian individuals with anxiety symptoms

Session 1	Psychoeducation regarding anxiety symptoms (e.g., cognitive, emotional, and somatic presentations of anxiety), and describe MABI sessions (e.g., components, structure, and homework expectation). The MABI theme for the first session included introducing the rules and practices, the raisin-eating exercise (mindful eating; eating raisins with the full involvement of the senses of smell, taste, sight and touch), and the body scan meditation (subjective evaluation of body parts, identifying stress and its relaxation), in which participants are instructed to notice sensations in each part of their body with curiosity and non-judgement. Link MABI techniques to spiritual practices through culture-specific metaphors. Persian language cards, handouts and worksheets were presented for each session.
Session 2	Show participants that how they see (or don't see) what is happening in their lives determines how they will react or respond. Body scan practice, pleasant event calendar (writing daily events that induced positive emotions such as joy, pride and praise), and being mindful in all actions (doing tasks with focus on them through the five senses). Review Persian language cards, handouts and worksheets about daily mindfulness practices.
Session 3	Focus on the theme pleasure and power of being in present moment. Individuals begin to explore how the immediate assignment of pleasant or unpleasant labels to experience is limiting. The body scan and sitting meditations are reinforced and mindful yoga, a practice of bringing non-judgmental attention to the moving of the body in simple standing and lying down postures, is introduced. Mindful watching and hearing (focusing of watching and hearing without judgment), sitting meditation, 3-min breathing space, mindful walking, and unpleasant event calendar (writing daily events that induced negative emotions such as distress, anxiety, sad, hate, anger, etc.). Integrate cultural syndromes, idioms of distress and anxiety relevant to Iranian populations. Discuss concerns about "distress" or "*parishani*," and "anxiety" or "*ezterab*". Review Persian language cards, handouts and worksheets about positive and negative emotions, regulating emotions through body scan and mindfulness practices.
Session 4	Focus on mindful watching and hearing and define the physiological and psychological bases of the anxiety reaction (fight/flight, or freeze) that helps participants understand the coping mechanisms involved with the short- and long-term effects of stress on the body and mind. Body scan and mindful yoga are reinforced; sitting meditation expands its focus from awareness of breath to include sensations in the whole body. Discuss escape and avoidance in maintaining symptoms. Identify avoidance patterns. Encourage use of adaptive coping skills and "facing up" to anxious thoughts and feelings in earlier sessions.
Session 5	Explore finding the space for making choices. Identifying and accepting unpleasant experiences, discussing insights and communication, sitting meditation, breathing space mindful yoga practice is reinforced; sitting meditation includes expanding awareness beyond breath, body sensations to hearing, to observing thoughts, emotions, and whatever arises in the present moment. Expanding cognitive flexibility and acceptance using culturally relevant discussions and culture-specific metaphors and local concepts.
Session 6	Focus on working with difficult situations, including stressful communication, and experimenting with and engaging in dialogue about recognizing their automatic patterns of relating and finding options to those patterns in stressful situations, while staying centered in body, mind, and emotions. There is formal teaching on mindful communication and exercises in mindful communication to explore bringing mindfulness to interpersonal relationships. Walking meditation-attending to the body's movement and the environment when walking-is introduced. This practice offers the opportunity once again to engage mindfully in an activity often done automatically without awareness
Session 7	Deepening self-knowledge and offering opportunities for insight into the impermanence of pleasant and unpleasant body-mind-states and two new practices are introduced. One new practice, a "mountain" meditation, uses guided imagery of the sitting body as a mountain, able to sit in stillness with stability and majesty through the storms and glories of days and years. Sitting like a mountain through all kinds of changing weather patterns and changing seasons is a metaphor for mindfulness meditation practice and life. The second new practice, loving-kindness meditation, guides participants in offering well-wishes to themselves, their loved ones, and increasingly wider circles of people, to include all beings. This meditation helps people recognize the inherent human qualities of friendliness, compassion, and connection that are available and capable of being cultivated.

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Session 8	Focus on integrating mindfulness practice more fully and personally into daily life. The learnings from this session include how to cultivate a disposition of generosity and compassion in formal meditation practice so that it may arise more readily in our day-to-day life.
Session 9	All meditations from prior seasons are engaged in again. Request participants write an autobiography. Acknowledge gains and discuss how skills and strategies can be integrated into daily life. Develop plans for maintaining gains and preventing relapse.

Content derived from Kabat-Zinn (2013), and Hayes et al. (2012).

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